

Data protocol

Official splits:

UNSW-NB15,
NSL-KDD

Cross-day:

CIC-IDS-2017,
CIC-DDoS2019

Training-only

Imputation

Integer encoding
SMOTE(-NC)

Tomek cleaning
One-hot expansion

Weight ablation

No weights

Original weights
Post-cleaning
residual weights

Model fitting

XGBoost

Five strong
baselines

MLP extension
(three datasets)

Evaluation

Macro PR-AUC

1% FPR budget

Paired bootstrap

TreeSHAP stability

Validation selects stopping and the 1% FPR threshold.
Locked test records produce every reported result.